

BS 1387:1985 A1, LIGHT, MEDIUM AND HEAVY

Description	BS 1387:1985 welded steel tube in produced is four thickness classes* - A1, Light, Medium and Heavy- and available in black finish or hot dipped galvanized finish in 6 meter uniform mill lengths.		
Applications	For ordinary conveyance of steam, gas, air, water, etc.		
End Finish and End Protection	Plain-end square-cut (PE) or threaded and fixed with coupling (T/C). (PE) tubes are shipped without any protection on both ends. T/C tubes are supplied screwed with taper threads to BS 21 and fitted with one taper-threaded malleable iron socket, as required under this specifications.		
Identification Marking	Tubes are marked by colour bands about 50 mm wide, painted about 300 mm from each end, as follows: Light tubes - Brown Medium tubes - Blue Heavy tubes - Red		

Chemical Composition	The chemical composition of the steel, by ladle analysis, shall comply with the table below:			
	C max.	Mn max.	P max.	S max.
	0.20%	1.20%	0.045%	0.045%

The mechanical properties at room temperature shall be as given in table below:

Yield strength (N/mm²) : 195 min.

Elongation on gauge length $L_0 = 5.65\sqrt{S_0}$ (%) : 20 min.

Outside Diameter : As shown in Table on opposite page.

Wail thickness	: Light tubes	-8%
	Medium and Heavy tubes	-10%

Mass : The mean consignment mass for quantities of 150 m and over of one size shall not deviate by more than $\pm 4\%$ from the mass of consignment calculated from the mass given in Table as appropriate. No single tube shall deviate by more than +10%, -8% from the mass given in Table as appropriate.

After the four successive one-minute immersions in the copper-sulphate solution, the test sample shall not show any adherent red deposits of metallic copper.

Black tubes up to and including DN 50 shall be bent cold without any signs of fracture or failure, through 180° round a former having a radius at the bottom of the groove equal to six times the outside diameter of the tube as given in Table. Hot-dip-zinc coated tubes shall be bent cold without cracking of the steel, through 90° round a former having a radius at the bottom of the groove equal to eight times the outside diameter of tube.

The flattening test applies to tubes greater than DN 50.

A ring not less than 40 mm in length taken from one end of each selected tube shall be flattened cold between parallel flat platens without showing either crack or flaw until the distance between the platens, measured under load, is not greater than 75% of original outside diameter of the tube, and no cracks or flaws in the metal elsewhere than in the weld shall occur until the distance between the platens is less than 60% of original diameter. The weld shall be placed at 90° to the direction of flattening.

The test shall be either a hydraulic test at a pressure of 50 bar ($50 \times 10^5 \text{ N/M}^2$), or alternatively, an eddy current test.

Hot-dip zinc coated tubes up to and including DN 25 shall have a rod 230 mm in length, of the appropriate diameter specified below, passed through them and shall have a free bore.

Rod Diameters	
Nominal Size (DN)	Diameter of rod (mm)
15.00	9.50
20.00	14.30
25.00	20.60

Note: *Special tube sizes not stipulated in the BS specification may be made available upon request. Please feel free to inquire.